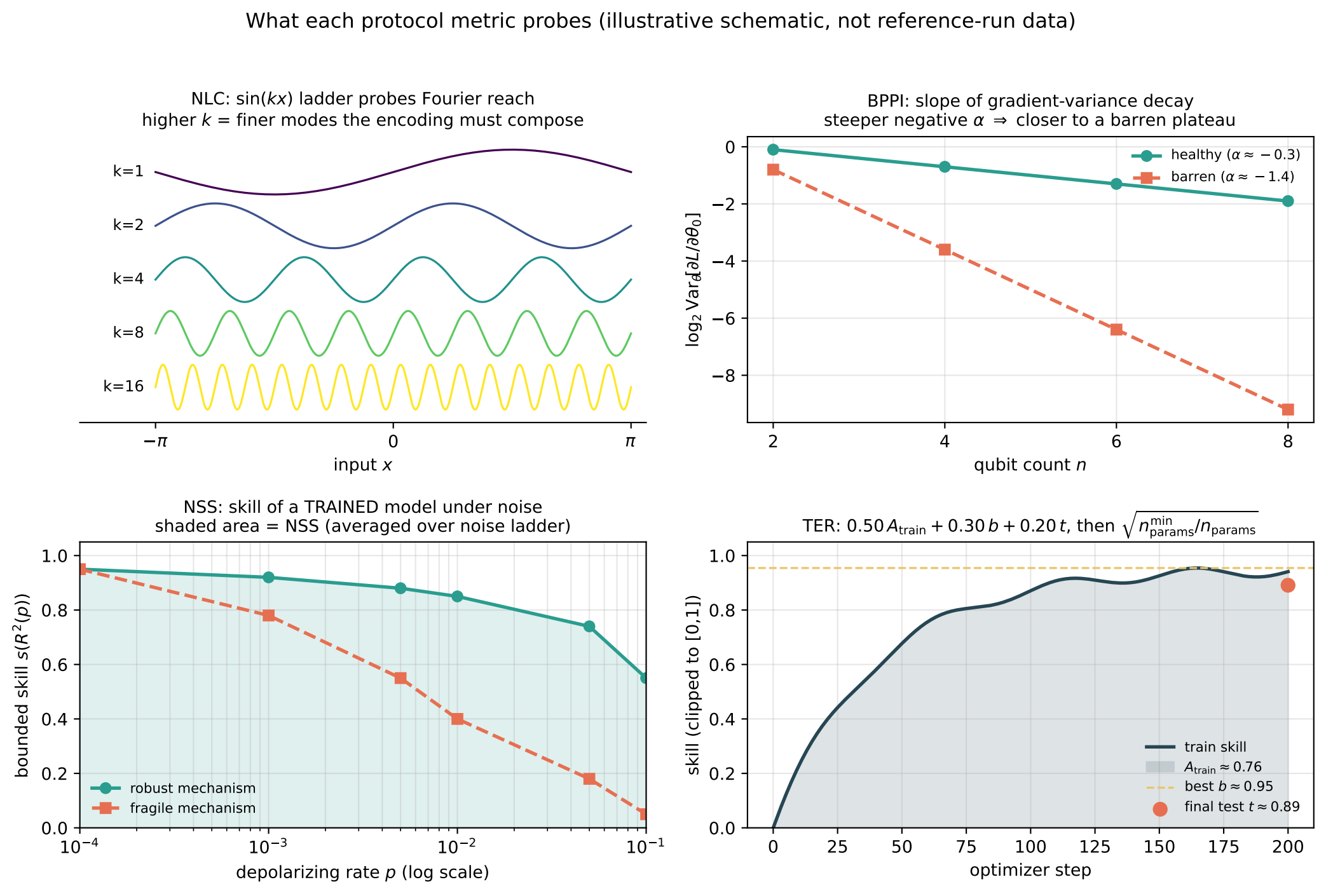
What each protocol metric probes (illustrative schematic, not reference-run data)
NLC: sin(kx) ladder probes Fourier reach
higher k = finer modes the encoding must compose
k=1
k=2
k=4
k=8
k=16
−π
0
π
input x
BPPI: slope of gradient-variance decay
steeper negative α ⇒ closer to a barren plateau
log2 Var_θ[∂L/∂θ0]
healthy (α ≈ −0.3)
barren (α ≈ −1.4)
qubit count n
NSS: skill of a TRAINED model under noise
shaded area = NSS (averaged over noise ladder)
bounded skill s(R²(p))
robust mechanism
fragile mechanism
depolarizing rate p (log scale)
TER: 0.50 A_train + 0.30 b + 0.20 t, then √(n_params^min / n_params)
skill (clipped to [0,1])
train skill
A_train ≈ 0.76
best b ≈ 0.95
final test t ≈ 0.89
optimizer step